

Negative Exponents with Variable Bases*Question Bank 3.T2*

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression using only positive exponents. Assume no variable equals 0.

1. a^{-6}

2. $\frac{1}{k^{-3}}$

3. $7y^{-4}$

4. $x^{-2}y^5$

5. $\frac{m^4}{n^{-3}}$

6. $\frac{2p^{-5}q}{r^{-2}}$

For questions 7 and 8, apply the laws of exponents. Write each answer using only positive exponents.

7. Write an equivalent expression.

a. $x^3 \cdot x^{-8}$

b. $(3y^{-2})^3$

c. $(2a^{-3}b^2)(5a^7b^{-6})$

8. Write an equivalent expression.

a. $\frac{m^2}{m^9}$

b. $\frac{12x^3y^{-2}}{4x^5y}$

c. $\left(\frac{a^3}{4b^{-7}}\right)^5$

Working backwards.

9. Find the value of n that makes each statement true.

a. $x^{-5} \cdot x^n = x^2$

b. $\frac{y^n}{y^4} = \frac{1}{y^6}$

c. $(m^n)^3 = \frac{1}{m^{12}}$

10. Let $x = 3$.

a. Find the value of x^2 and the value of x^{-2} .

b. Find the value of $-x^2$. Is it equal to x^{-2} ? Explain.

11. Which expression is equivalent to $5x^{-3}$?

- ☐ A. $\frac{1}{125x^3}$
- ☐ B. $\frac{5}{x^3}$
- ☐ C. $\frac{1}{5x^3}$
- ☐ D. $-5x^3$

12. Select ALL of the expressions that are equivalent to $\frac{1}{x^6}$.

- ☐ A. x^{-6}
- ☐ B. $(x^{-2})^3$
- ☐ C. $\frac{x^3}{x^9}$
- ☐ D. $(x^{-3})^{-2}$
- ☐ E. $-x^6$
- ☐ F. $x^2 \cdot x^{-8}$